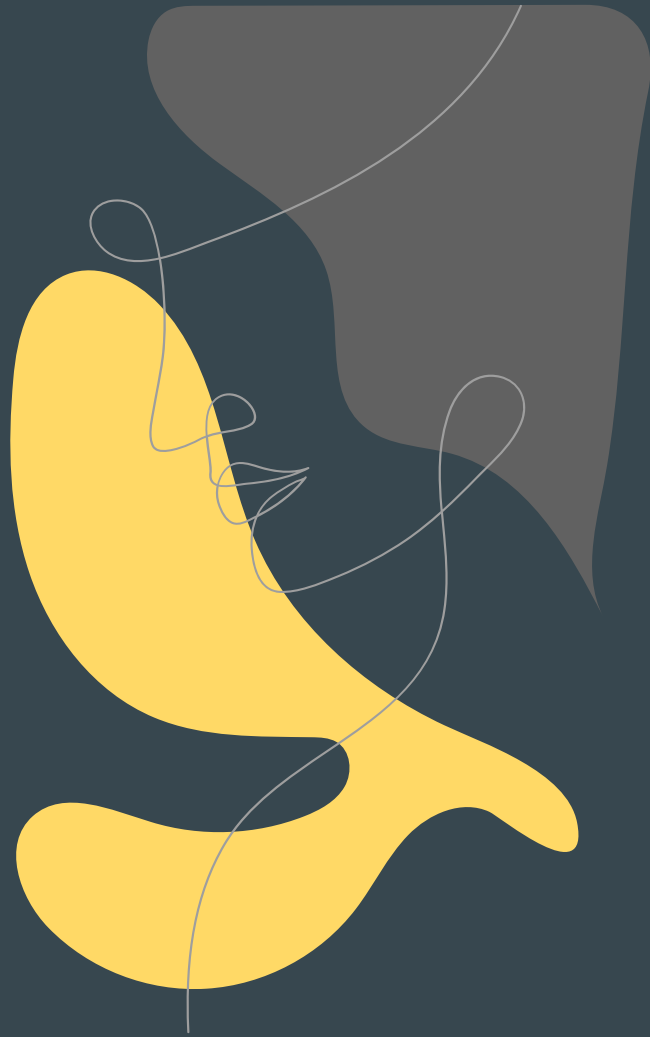


Congenital Prosopagnosia: impact on different types of recognition

The HemoGoblins: Anika Soin, Ava
Waite, Manjari Muruganandam, Megan
Puckett, Sean Stahl



What is face recognition?

- **Face perception**

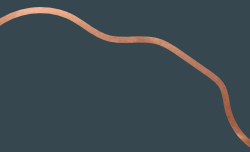
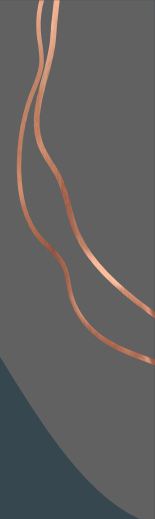
- Involves detecting a face in a visual scene (face detection)
- Involves discriminating a specific face from others (face discrimination)
- Includes associating something that is perceived with a stored representation of a specific face in memory (face recognition)

- **Social cognition importance**

- Facial expressions provide clues about feelings and reactions
- Supports the development of social abilities and interactions



What is **prosopagnosia**?

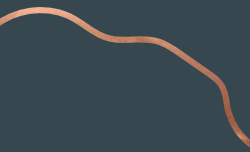
- **Face recognition deficit**
 - Damage or developmental failures affect neural areas involved in face recognition
 - **Acquired**
 - Loss of familiarity for facial identity
 - Typically caused by damage to occipitotemporal or anterior temporal areas, usually on the right or bilaterally
 - **Developmental or congenital**
 - Functional and structural origins still under investigation
 - Family history
- 
- 
- 



What do we know so far?

- Some studies propose the **fusiform face area** (FFA) as the most **face-selective** area
- FFA selectivity may be influenced by visual cues like color and spatial frequency
- **Lesion studies**
 - Suggest the lateral part of the right inferior occipital cortex as a critical area for prosopagnosia
 - Supported by **lesion overlap** and **electrical stimulation** studies

(Bouvier & Engel, 2006), (Jonas et al., 2012)



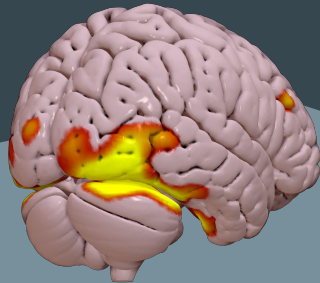
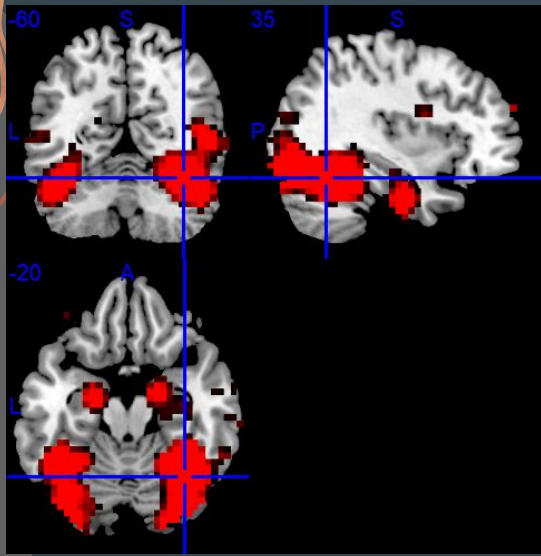


Research questions:

Are there shared neural substrates underlying different types of recognition (i.e. face and object)?

How are these substrates impacted by the impaired face recognition abilities in congenital prosopagnosia?

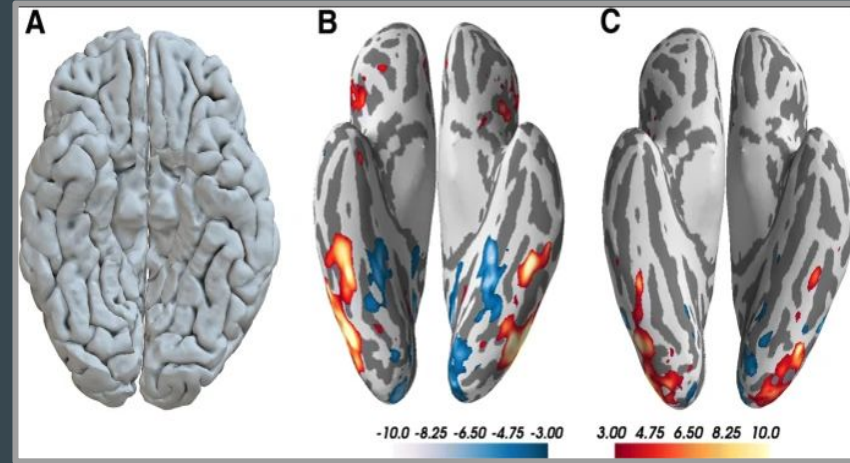
NeuroQuery visualizations

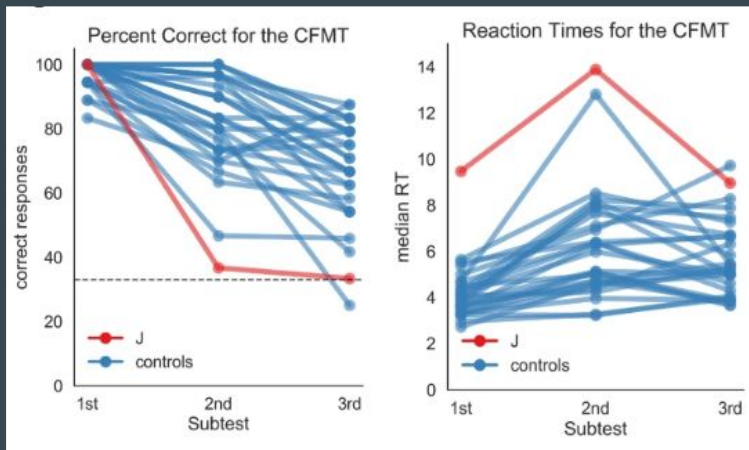


- “Congenital prosopagnosia”
- Top: **MRICron**
 - Surface-based visualization
 - Activation of hippocampus in temporal lobe is clearer in the sagittal surface view
- Bottom: **SurfIce**
 - Volume-based visualization
 - Shows more general patterns of activation
- Network areas
 - “Core” = FFA, OFA, superior temporal sulcus
 - “Extended” = anterior temporal face area, inferior frontal gyrus, precuneus

The Hidden Identity of Faces: A Case of Lifelong Prosopagnosia

- Subject J has **congenital prosopagnosia**
- Adapted through type of voice, gait, and clothes
- **Univariate** analysis:
 - Deficits in learning new faces/recognizing familiar faces when only inner features available
 - Recognition intact with uncropped faces and blurred features
 - fMRI tasks showed prototypical activity in **lateral fusiform gyrus**
 - Recognition issues not due to abnormal brain structure/function in regions





What can be concluded:

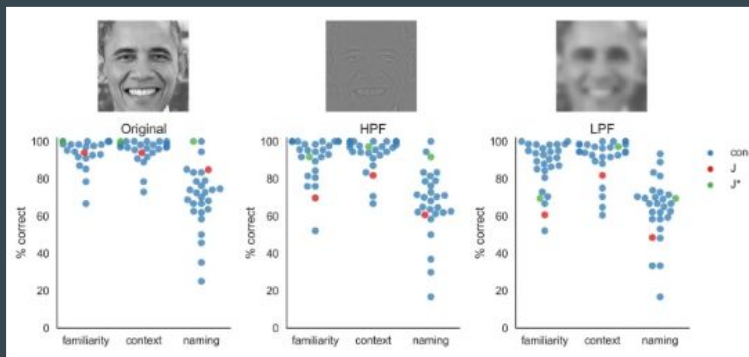
Severe deficits in recognizing faces, especially based on inner features.

Differentiates between neural activation and cognitive processes in face recognition.

What cannot be concluded:

The specifics of cognitive dysfunction in natural, real-world face recognition scenarios.

Whether findings can be generalized across all cases of CP.



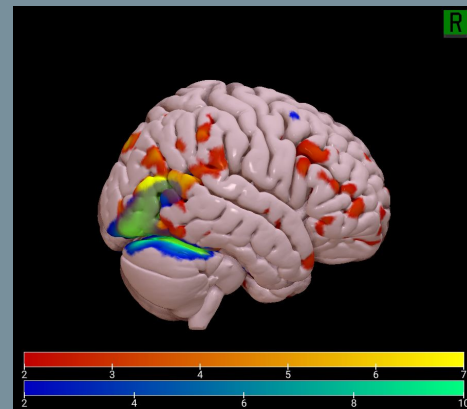
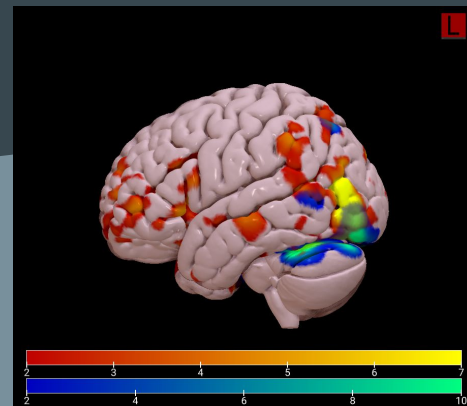
Limitations:

Single-case study limits generalizability.



Visualization of T-maps

- Overlap of “**faces vs landscapes** ” from neurovault case study and NeuroQuery search for “congenital prosopagnosia”
- Overlap in **fusiform face area** (inferior temporal lobe)
- Study’s T-map shows brain activity in **frontal** and **parietal** lobes
- Differences likely due to other tasks asked of patient in congenital prosopagnosia study
 - **Landscape** recognition



Controls

CPs

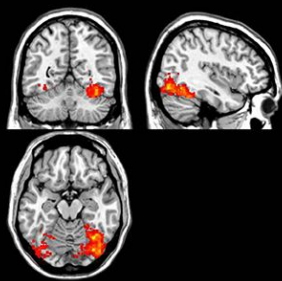
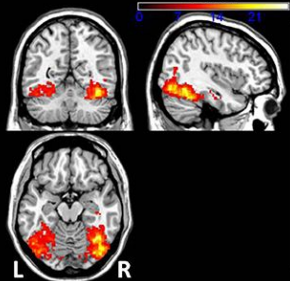
t-value

0 1 2 3 4 5

A



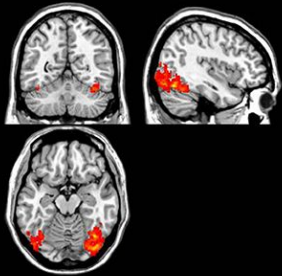
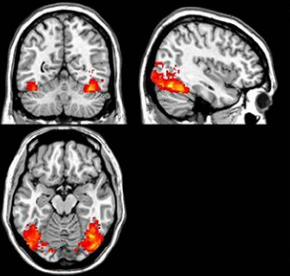
Vs.



B



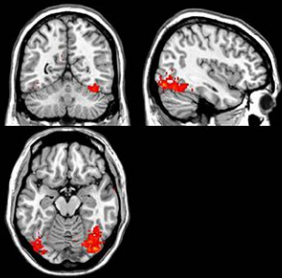
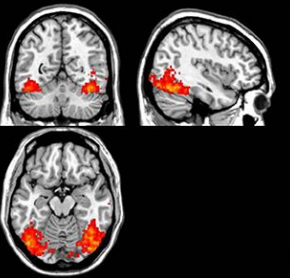
Vs.



C



Vs.



MVPA Reveals Abnormal fMRI Activity in the “Core” and “Extended” Face Network

- fMRI and MVPA used to investigate neural responses to faces, objects, bodies, and body parts in CP individuals
- **Univariate** statistics failed to report reduced discriminability between faces and objects picked up by **multivariate** analyses
- **MVPA** more sensitive approach in detecting group differences
- **Limitation:** potential variability in location of face-sensitive regions between subjects



What is next?

Population: within-subjects design (CP) and between-subjects design (CP and controls)

Tasks: face recognition, object recognition, voice recognition

Behavior: accuracy, reaction time, subjective ratings of confidence or difficulty during recognition tasks

Additional measures: EEG alongside fMRI in order to record neural activity with high temporal resolution during recognition

Analysis approach: multivariate analysis (MVPA) in order to examine **connectivity** across brain regions and **representational similarity** underlying different types of recognition

Circuits: FFA (face-specific processing), IT (inferotemporal cortex for object recognition), right temporal lobe for voice-identity recognition



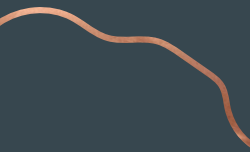


Overall hypothesis:

Shared neural substrates underlie different types of recognition.

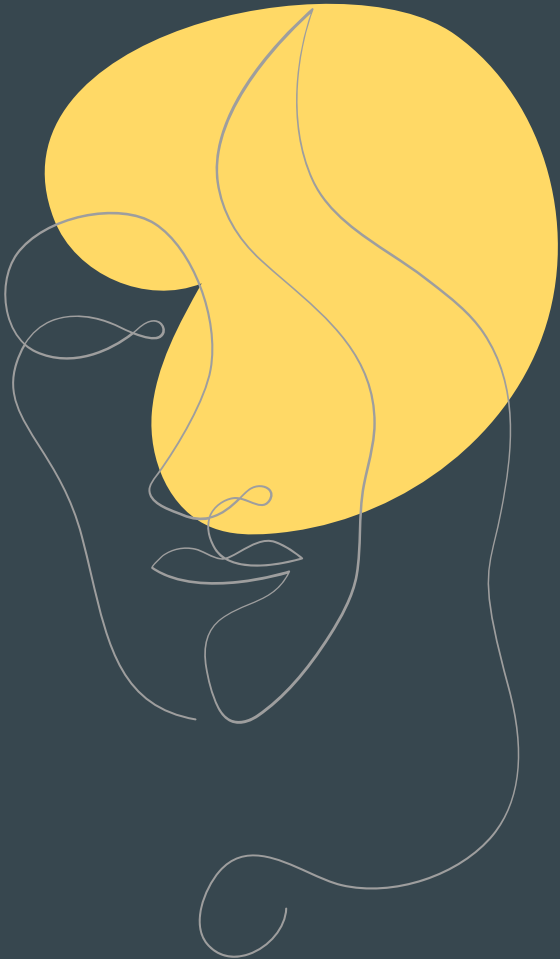
Impaired face recognition in CP is associated with distinct neural patterns compared to controls, potentially involving compensatory mechanisms in non-face recognition areas.

We expect that other types of visual discrimination may be compromised, however non-visual-dependent types of recognition will not be negatively affected.



In conclusion,

- Studies reveal possible compensatory mechanisms in non-face recognition areas, emphasizing the distinct neural patterns associated with impaired face recognition in CP
- Further exploration into the impact on visual-dependent recognition in CP needs to be conducted
- **Interesting fact:**
 - According to Cortex (February 2023), **1 in 33** people may meet the criteria for prosopagnosia
 - More than **10 million** Americans



**Thank You For
Listening!**



Sources:

[Face Perception](#)

[Prosopagnosia: Current Perspectives](#)

[NeuroQuery](#)

[MRICron](#)

[Surf Ice](#)

[The Hidden Identity of Faces: A Case of Lifelong Prosopagnosia](#)

[NeuroVault](#)

[MVPA Reveals Abnormal fMRI Activity in Both the "Core" and "Extended"](#)

[Face Network in Congenital Prosopagnosia](#)

[How Common Is Face Blindness?](#)

